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CONTENT OF APPENDIX

In this paper, we introduce SOTOPIA to encourage research on interactive social intelligence. We showed that SOTOPIA can be used for evaluating social interaction among models and humans. In the appendix, we provide the following items that shed further insight into these contributions:

- A Extended related work;
- B The limitations of SOTOPIA and future directions;
- C formal definition of SOTOPIA from a multi-agent reinforcement learning perspective and technical details of generating social tasks;
- D the prompt we use for GPT-4 (OpenAI, 2023) to evaluate model performance;
- E The Amazon Mechanical Turk interface for evaluating model performance;
- F The procedure and interface for humans¹³ when playing characters in SOTOPIA;
- G Additional quantitative results;
- H Additional qualitative examples.

A EXTENDED RELATED WORK

There have been a lot of social science works that have done agent-based modeling to study human interactions, spanning across various domains such as economics, psychology, and education (Sawyer, 2005; Rosé et al., 2008; Deguchi, 1995). Prior simulation environments have played a pivotal role in constructing theories and generating hypotheses in these fields. However, they frequently constrain agents’ communicative capacities to artificial languages and present a highly reductionist view of simulated human behavior (Gilbert, 2005; Tesfatsion & Judd, 2006; Huang et al., 2014; Kovač et al., 2021; Urbanek et al., 2019). LLMs provide a more flexible and expressive way to model human behavior. Here, we include a more detailed discussion of the recent works investigating LLMs for simulating human social interactions. There are works that focus on investigating the fidelity of LLMs in keeping the designated persona and experiences of the characters (Shao et al., 2023; Jiang et al., 2023). There are works that simulate human social interactions focusing on certain aspects such as competition, collaboration, negotiation, deception, problem-solving and etc., (Zhang et al., 2024; Zhao et al., 2023; Liu et al., 2023; Michael et al., 2023; Rasal, 2024; Hubinger et al., 2024; Bianchi et al., 2024; Xie et al., 2024; Jiang et al., 2023). As LLMs are becoming more and more popular in simulating human social interactions, there are also works that focus on investigating the potential issues and challenges of using LLMs in social simulations, such as stereotypes and reporting issues (Cheng et al., 2023b; Zhou et al., 2024).

B LIMITATIONS & FUTURE DIRECTIONS

We identify SOTOPIA as the first platform for a general and realistic evaluation of social intelligence in AI agents. To better understand the social intelligence of AI agents, we discuss some future directions for SOTOPIA and the field of AI social intelligence.

Limitations of the simplified simulated “world” As every simulation is a simplification of the real world, SOTOPIA identifies several key components of realistic social interactions, while abstracting aspects of the real world. First, we consider five types of social relationships in SOTOPIA. Future work could expand the type and granularity of social relationships (e.g., colleagues, classmates, etc.) in SOTOPIA. Different types of relationships would require agents to exhibit different social behaviors (Jenkins et al., 2018), making the expansion of relationship types an important future research direction. Second, future work could expand the breadth of the character and social scenario pool in SOTOPIA to cover more social behaviors. Third, SOTOPIA constrains the fixed turn-taking interaction to the *dyadic context*, studying interactions between two agents. Future works could tackle more complex social interactions, such as multi-party interactions and those involving complex dynamics (e.g. asynchronous interactions, interruptions).

¹³All the human subjects experiments are approved by the Institutional Review Board (IRB) at the authors’ institution.